

AYE
TECH HUB

• SOLAR ENERGY • LESSON 05

Solar Batteries

Storage, Sizing and Selection

Lesson 05 of 30 | Solar Energy Program

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- Lead-acid vs Lithium (LFP/NMC) — full chemistry comparison
- State of Charge, Depth of Discharge, C-rate explained
- Step-by-step battery bank sizing calculation
- Series and parallel battery bank wiring rules
- Which battery is right for your system — decision guide

Battery Chemistry — The Foundation of Your Choice

The battery is often the most expensive component in an off-grid or hybrid solar system and determines how long your system will work reliably. The choice of chemistry affects every aspect of the system — cost, lifespan, safety, how much energy you can actually use, and how much maintenance is required.



Figure 1 — Battery Chemistry Scorecard (10 = best)

Chemistry	Nominal Voltage	Typical DoD	Cycle Life	Cost / kWh	Key Advantage
Flooded Lead-Acid (FLA)	2V/cell (12V = 6 cells)	50%	300–600 cycles	Lowest (\$80–120)	Cheapest upfront; widely available; robust
AGM (Sealed Lead-Acid)	12V or 6V monobloc	50%	400–700 cycles	Low (\$120–180)	Maintenance-free; no venting needed; spill-proof
Gel Lead-Acid	12V or 6V monobloc	50%	500–1000 cycles	Medium (\$140–200)	Good deep-cycle; tolerates heat; slow charge OK
LFP — Lithium Iron Phosphate	3.2V/cell (12.8V = 4 cells)	80–90%	3000–6000+ cycles	High (\$300–500)	Safest lithium; longest life; best for solar storage
NMC — Nickel Manganese Cobalt	3.6V/cell (14.4V = 4 cells)	80%	1500–3000 cycles	High (\$350–600)	Higher energy density; used in EV-based home batteries

Why LFP dominates solar storage in 2026: LFP cycle life of 4000+ cycles at 80% DoD equals 11+ years at one full cycle per day. Lead-acid at 50% DoD and 500 cycles lasts under 2 years with daily cycling. Despite higher upfront cost, LFP cost per kWh over lifetime is 3–5× cheaper than lead-acid in solar applications.

Key Battery Concepts Every Solar Engineer Must Know

Term	Symbol	Definition	Practical Example
State of Charge	SoC (%)	The current charge level as a percentage of full capacity. 100% = fully charged. 0% = fully discharged (never reach in practice).	SoC 80% on a 200Ah battery = 160Ah available
Depth of Discharge	DoD (%)	How deeply the battery is discharged. DoD = 100% – SoC. Higher DoD means more usable energy but reduces cycle life.	Lead-acid max DoD: 50%. LFP max DoD: 80-90%.
Usable Capacity	C_usable (Ah or kWh)	Actual energy you can draw. C_usable = C_nominal × DoD.	200Ah LFP at 80% DoD → 160Ah usable 200Ah FLA at 50% DoD → 100Ah usable
C-Rate	C or xC	Charge/discharge rate relative to capacity. 1C = full charge/discharge in 1 hour. C/10 = 10 hours. 2C = 30 minutes.	LFP: discharge at 1C (normal) or 2C (short burst) FLA: max discharge 0.2C
Self-Discharge	% per month	Energy lost when battery sits unused. Higher = more energy wasted in storage.	LFP: 1-3%/month. FLA: 5-15%/month. NMC: 1-5%/month
Battery Efficiency	η (%)	Round-trip efficiency: energy out ÷ energy in × 100%. The rest is lost as heat during charge/discharge.	LFP: 95-99%. Lead-acid: 70-85%.
Calendar Life	Years	Total lifespan regardless of cycles. Even unused batteries degrade. Temperature is the biggest factor.	LFP: 10-15 years. FLA: 3-5 years. NMC: 8-12 years.

SoC, DoD and Usable Energy — Visualised

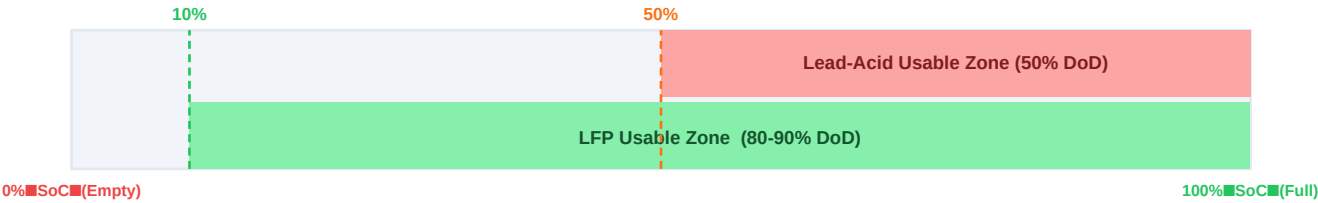


Figure 2 — State of Charge vs Usable DoD: LFP vs Lead-Acid

Temperature warning: Battery capacity and life are heavily temperature-dependent. Lead-acid loses ~1% capacity per °C below 25°C. LFP charging below 0°C causes lithium plating — permanent damage. Always derate capacity and protect batteries from extremes: keep between 10°C and 35°C for best performance and life.

Battery Bank Sizing and Wiring

Battery bank sizing is one of the most critical calculations in off-grid solar design. Undersize and your loads run out of power. Oversize and you waste money on unused capacity. The goal is to supply your required energy for the desired autonomy days without exceeding the battery's maximum DoD.

Step	Formula	Example (Daily load 3 kWh, 2 days autonomy, LFP 48V system)
1. Total energy needed	$E_{total} = \text{Daily load (kWh)} \times \text{Autonomy days}$	$E_{total} = 3 \text{ kWh} \times 2 \text{ days} = 6 \text{ kWh}$
2. Account for battery efficiency	$E_{adj} = E_{total} \div \text{Battery efficiency}$	$E_{adj} = 6 \text{ kWh} \div 0.97 = 6.19 \text{ kWh (LFP 97\%)}$
3. Apply DoD limit	$E_{nominal} = E_{adj} \div \text{DoD}$	$E_{nominal} = 6.19 \div 0.80 = 7.74 \text{ kWh minimum bank}$
4. Add temperature derating	$E_{derated} = E_{nominal} \div \text{Temp_factor}$ (0.85 for 40°C ambient)	$E_{derated} = 7.74 \div 0.85 = 9.1 \text{ kWh} \rightarrow \text{round up to 10 kWh}$
5. Convert to Ah at system voltage	$C_{bank} \text{ (Ah)} = E_{derated} \text{ (Wh)} \div V_{system}$	$C_{bank} = 10\,000 \text{ Wh} \div 48\text{V} = 208 \text{ Ah}$
6. Select battery units	$N_{batteries} = C_{bank} \div C_{per_battery}$	$208 \text{ Ah} \div 100 \text{ Ah/battery} = 2.08 \rightarrow \text{use } 4 \times 100\text{Ah (2 series for 48V, 2 parallel for 200Ah)}$

Series and Parallel Battery Wiring

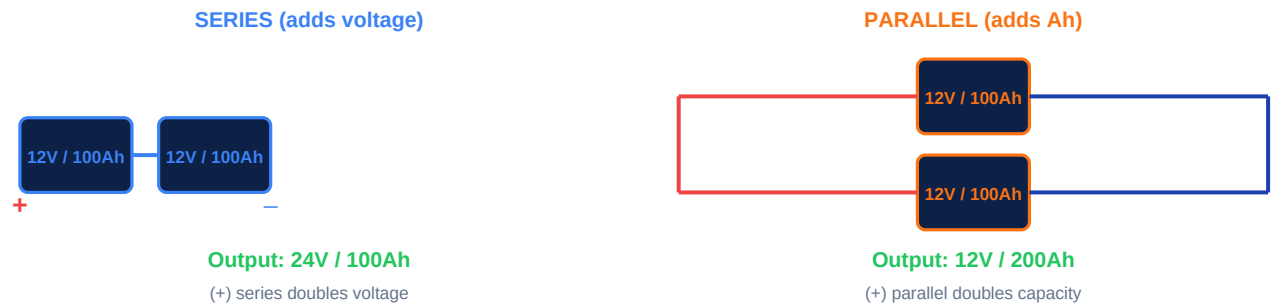


Figure 3 — Series vs Parallel Battery Bank Wiring

Wiring	Voltage	Capacity (Ah)	Rule
Series (+ to -)	Adds: $2 \times 12\text{V} = 24\text{V}$	Stays same: 100Ah	Connect positive of one to negative of next. Use identical batteries.

Parallel (+ to +, – to –)	Stays same: 12V	Adds: $2 \times 100\text{Ah} = 200\text{Ah}$	Connect all positives together, all negatives together. Fuse each battery.
Series-Parallel (2S2P)	$2 \times 12\text{V} = 24\text{V}$	$2 \times 100\text{Ah} = 200\text{Ah}$	Build series strings first, then connect strings in parallel. Match all strings.

Battery Selection Decision Guide

Your Situation	Recommended Chemistry	Why
Tight budget, infrequent cycling (holiday cabin)	AGM or Gel Lead-Acid	Lower upfront cost; manageable if not cycled daily
Daily cycling, 5+ year lifespan needed	LFP (Lithium Iron Phosphate)	Best cycle life per dollar over 5+ years; safest lithium
Warm climate ($>35^{\circ}\text{C}$ average temperature)	LFP with BMS temperature management	LFP tolerates heat better than lead-acid; BMS protects against overtemp
Cold climate ($<5^{\circ}\text{C}$ regular temperatures)	Heated LFP enclosure or AGM at reduced capacity	LFP must not charge below 0°C ; AGM handles cold better but with lower capacity
Large commercial / industrial system	LFP rack-mounted (e.g. Pylontech, BYD, CATL)	Scalable, BMS included, warranty, integration with hybrid inverters
Maximum energy density (space-limited)	NMC (e.g. Tesla Powerwall, LG RESU)	Higher Wh/kg than LFP; good choice when physical size is constrained

What comes next — SOLAR-006: Inverters Explained — string inverters, microinverters, hybrid inverters, off-grid inverters, MPPT tracking, inverter sizing, and how to match an inverter to your solar panels and battery bank.